CODECHECK certificate 2026-029

https://doi.org/10.5281/zenodo.23666248

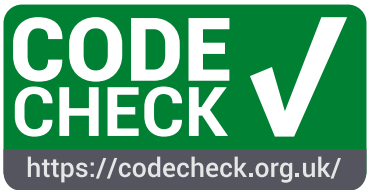


Table 1: CODECHECK summary

Item	Value
Title	*Do Laypeople Endorse the Axioms of Inequality Measurement? Outcomes Report of a Preregistered Replication and Extension of Amiel and Cowell (1992) in a U.S. General-Population Sample*
Author(s)	Cristóbal Moya, Carsten Schröder, Philipp Lersch
Reference	https://doi.org/10.71240/lcyc.829503
Repository	https://doi.org/10.17605/OSF.IO/8Z2UV
Codechecker(s)	Nataliia Petliak
Date of check	*2026-10-22*
Summary	The authors' computational reproduction workflow completed successfully and generated the expected PDF report. An initial Docker build issue was resolved by explicitly selecting the target platform.

Table 2: Summary of output files generated

File	Comment	Size (b)
`outcomes_report_pub.pdf`	Reproduced 25-page PDF report containing the analyses, tables, and figures. The complete report is included in this certificate.	634352

Version checked: `inequality-axioms-repro-20260831.zip`, version 1.

Summary

The authors' computational reproduction workflow completed successfully and generated the expected PDF report. An initial Docker build issue was resolved by explicitly selecting the target platform.

CODECHECKER notes

Computational environment

The CODECHECK was performed on an Apple Silicon Mac running macOS.

The certificate-generation environment uses Python 3.12.15, Jupyter, and Typst 0.15.1.

The Docker environment used R 4.6.0 and Quarto 1.10.18. According to Cristóbal Moya, the analysis used the 2026-06-23 R package snapshot from the `rocker/tidyverse:4.6.0` base image.

Docker 28.5.1 was installed and the Docker engine was running.

Reproduction procedure

The computational reproduction followed the four-step Docker workflow described in the authors' `README.md` .

Steps 1–2: Download and extract the reproduction package

The reproduction materials were downloaded from the authors' OSF repository and extracted into the local CODECHECK working directory.

The downloaded archive (`8z2uv-osfstorage-archive.zip`) contained a nested archive (`inequality-axioms-repro-20260831.zip`).

The extracted package includes the Quarto source document (`output/outcomes_report_pub.qmd`), minimal anonymized datasets under `data/minimal/` , a `Dockerfile` , a `Makefile` , and reproduction instructions. Separate licenses are provided for the code and data.

Extraction completed successfully. No modifications were made to the authors' source files or configuration.

Step 3: Build the Docker image

The initial Docker build was attempted from the root of the extracted reproduction package using the command specified in the README:

```
docker build -t ineq-axioms-repro:1.0 .
```

The build failed after approximately 62 seconds during the installation of Quarto 1.10.18.

The container was running on ARM64, but the Dockerfile attempted to install the AMD64 version of Quarto. This resulted in the following error:

```
dpkg: error processing archive /tmp/quarto.deb (--install):
package architecture (amd64) does not match system (arm64)
```

The Dockerfile specifies `ARG TARGETARCH=amd64` , which was not overridden during the initial build.

Consequently, the downloaded Quarto package did not match the container architecture.

The build terminated with exit code 1. The failure was caused by an architecture mismatch in the supplied Docker configuration when building on Apple Silicon.

To address the mismatch without modifying the authors' Dockerfile, the build was retried with the target platform explicitly set to AMD64:

```
docker build --platform linux/amd64 -t ineq-axioms-repro:1.0 .
```

This uses Docker's AMD64 emulation on the Apple Silicon host. The command differs from the authors' instructions only by the addition of `--platform linux/amd64`.

The second build completed successfully in 355.8 seconds, including installation of Quarto 1.10.18 and the required R packages. The resulting image was tagged `ineq-axioms-repro:1.0`.

No modifications to the authors' files were necessary.

Step 4: Render the report

After successfully building the Docker image for AMD64, the report was rendered using the exact command provided in the authors' README:

```
docker run --rm -v "$PWD":/work ineq-axioms-repro:1.0
```

Docker reported a platform mismatch warning because the AMD64 image was running on an ARM64 host:

```
WARNING: The requested image's platform (linux/amd64) does not match
the detected host platform (linux/arm64/v8) and no specific platform
was requested
```

The warning did not prevent execution.

Quarto completed all 89 processing steps, including the code chunks generating the analyses, tables, and figures. Pandoc and Typst then compiled the document successfully.

The process completed without errors and generated the expected output:

```
output/outcomes_report_pub.pdf
```

The authors' source files and configuration remained unchanged.

Results and observations

The reproduction workflow completed successfully using the authors' Docker environment with AMD64 emulation on the Apple Silicon host. The expected PDF was generated without modifying the authors' source files.

The reproduced PDF was visually inspected, and the tables and figures appeared to render correctly. A direct comparison with the published figures was not possible because they were not loading on the Lifecycle Journal website at the time of the check.

I did not perform a detailed numerical comparison with the published report.

Cristóbal Moya compared the reproduced PDF with his own render. He found that all estimates, confidence intervals, sample sizes, percentages, and figures matched. For 26 p-values below 1e-8, the two renders differed in the third or fourth significant digit. The reproduced PDF was 25 pages and his was 26. He attributed the difference in page count to a different version of the table package.

Recommendations to the authors

In my initial build on Apple Silicon, the Dockerfile in the checked archive tried to install the AMD64 version of Quarto in an ARM64 container, causing the build to fail.

Consider updating the Dockerfile to support multiple architectures or documenting the need to build with `--platform linux/amd64` on ARM64 hosts.

Cristóbal Moya reported that the authors have updated their Dockerfiles to install the Quarto build that matches the machine's architecture. He said a corrected release of this reproduction package would include that change.

It would also be helpful to export each figure as a separate graphics file, in addition to the complete PDF report. This would make it easier to inspect individual figures, trace them back to the code that generated them, and include them explicitly in the reproduction manifest.

Manifest files

Summary of all reproduced output files (tables, text files, figures, …).

`output/outcomes_report_pub.pdf`

Comment: *Reproduced 25-page PDF report containing the analyses, tables, and figures. The complete report is included in this certificate.*

Do Laypeople Endorse the Axioms of Inequality Measurement?

Outcomes Report of a preregistered replication and extension of Amiel and Cowell (1992) in a U.S. general-population sample

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Abstract

The transfer principle is foundational to inequality measurement: a rank-preserving transfer of income from a richer to a poorer person must reduce inequality. Yet Amiel & Cowell (1992) found that only 35% of surveyed students endorsed it in numeric form. We test whether this finding generalizes to the U.S. adult population (H1), whether the transfer principle is the least supported of five core axioms (H2), and whether endorsing it predicts redistribution preferences (H3) or inequality aversion (H4). In a preregistered study (OSF registration 10.17605/OSF.IO/82DJS), 3,695 respondents completed the original questionnaire. Two experimental treatments—proportional visual aids and a performance bonus—tested whether low measured support is an artifact of presentation or effort. A robustness sample (N = 2,125) applies the registered attention-check and speeding exclusions. Tests comprised one-tailed proportion tests (H1), a logistic mixed model over axioms with respondent random intercepts (H2), and regressions of redistribution preferences, tolerated efficiency losses in a leaky-bucket task, and incentivized donation on transfer-principle support (H3, H4).

Numeric transfer-principle support was 26.3% (95% CI [24.9%, 27.8%]) and verbal support 30.0%, both below H1′s 66% threshold; all four other axioms drew significantly more support (numeric-format support 32.1%–55.9%), so H1 and H2 were supported. Transfer-principle support did not predict public and private redistribution preferences, tolerated efficiency losses, or donation of an earned bonus; H3 and H4 were not supported. In the case of redistribution preferences and efficiency-loss outcomes, the confidence intervals exclude positive differences larger than 0.26 scale points (1–7) and 0.05 (0–1 index). The same pattern holds in the robustness sample. Low support for the transfer principle generalizes beyond student samples, but endorsement predicts neither redistributive preferences nor behavior.

Keywords: income inequality; transfer principle; inequality axioms; redistribution preferences; inequality aversion; leaky bucket; replication.

1 Introduction

Virtually every applied measure of income inequality is built on a small set of axioms. Chief among them is the transfer principle: a rank-preserving transfer of income from a richer to a poorer person must reduce inequality. Also known as the Pigou–Dalton principle (Pigou, 1912), it is what makes the Lorenz ordering, the Gini coefficient, and the family of relative inequality indices behave the way analysts expect (Clark & D'Ambrosio, 2015).

Amiel and Cowell (1992) subjected this and four other core axioms (scale invariance, translation invariance, the population or replication principle, and decomposability) to an empirical test outside the expert community. They asked university students in the United States, Germany, the United Kingdom, and Israel to compare pairs of income distributions, once in numerical and once in verbal form. Their results were striking: only 35% endorsed the transfer principle in the numerical comparison, and roughly 60% in the verbal statement. Support for the other axioms was similarly limited. If the axioms that experts treat as self-evident are endorsed by only a minority of respondents, there is a substantive gap between the foundations of inequality measurement and the public understanding of inequality.

We ask whether that divergence between expert axioms and lay perceptions holds in a general adult population, and whether endorsing the transfer principle carries any broader distributive meaning in the public's understanding of inequality. Four considerations motivate these questions. First, the original

finding could be an artifact of sample composition: economics students may differ from the general population in their preferences regarding distribution and redistribution, potentially as a result of their educational background. Second, the finding could be specific to its period: public debate about economic inequality has changed markedly since the early 1990s, and endorsement of measurement principles in the 2020s need not match what students expressed then. Third, the relative pattern originally found speaks directly to debates about reducing inequality through redistribution and about the equality–efficiency trade-off: is the transfer principle the least endorsed of the five axioms in the general population? Fourth, and beyond replication: does endorsing the transfer principle influence preferences about inequality? If agreement with it reflects an underlying egalitarian orientation, supporters should also express stronger redistribution preferences and greater inequality aversion. If it does not, agreement may reflect task-specific comprehension unrelated to distributive dispositions.

Here we report a preregistered replication and extension of Amiel and Cowell (1992) addressing these questions in a large U.S. general-population sample. A large-scale general-population replication adds more than a check on the original result: it improves external validity and generalizability beyond student convenience samples, and its statistical precision supports population-level estimates and heterogeneity analyses — advantages parallel to those of recent large-scale replications of influential small-sample findings in experimental philosophy, such as the Knobe effect (Knobe, 2003; Maćkiewicz et al., 2024; Schmidt-Petri et al., 2026; Sytsma et al., 2022). Alongside the direct replication, two experimental treatments test whether low measured support is an artifact of presentation or effort: one adds proportional visual representations of the compared distributions, the other pays a bonus of up to $20 for thoughtful and consistent answers.

The hypotheses connect these considerations to registered tests. H1 re-tests the original finding against the threshold that Amiel and Cowell treated as the marker of a surprising result. H2 concerns the relative pattern: in the original numeric results the transfer principle was among the least supported axioms, and Amiel and Cowell singled it out as the sharpest point of lay–expert divergence; we test whether it is the least supported of the five in a general population. H3 and H4 test whether endorsement carries distributive meaning. If agreeing with the transfer principle expresses an egalitarian orientation, supporters should more strongly favor redistribution by public and private actors (H3), and should reveal greater inequality aversion where it has a price: tolerating larger efficiency losses in a leaky-bucket task, in which only part of what is taken from richer people reaches poorer people, and donating an earned bonus to an organization working to reduce inequality (H4). If endorsement instead reflects task-specific comprehension, none of these associations should appear. In a nutshell, this paper tests the following hypotheses:

- **H1.** Support for the transfer principle is highly contentious.
- **H2.** The transfer principle is the least supported axiom, relative to scale invariance, translation invariance, replication invariance, and decomposability, when asked in numeric form.
- **H3.** Stronger support for the transfer principle is associated with stronger preferences for redistribution by (a) public and (b) private actors.
- **H4.** Stronger support for the transfer principle is associated with greater inequality aversion: (a) accepting less efficient transfers, and
  (b) donating to reduce inequality.

These hypotheses, the sampling plan, and the analysis plan were registered before data collection (Moya et al., 2025), OSF registration 10.17605/OSF.IO/82DJS. This report follows the registered Study Design Table available in the registration to conduct the confirmatory tests and their prospective interpretation.

2 Data and Methods

2.1 Design

We used a survey administered in computer-assisted self-interviewing (CASI) mode with a between-subjects experimental design with three arms. All three arms administer the same replication core—the original Amiel–Cowell items in their original wording and format—so every respondent contributes to the confirmatory replication tests. The arms differ only in presentation and incentives. The Original Questionnaire Control arm (planned $n \approx 1{,}500$) presented the items with no visual aids and no performance incentive. The Visual Cues arm (planned $n \approx 1{,}500$) presented the same items, adding proportional visual representations of the two income distributions compared in each of the nine numeric comparison ques-

tions (Q1–Q9; see Measures). The Incentivized arm (planned $n \approx 300$) presented the original items, with respondents told they could earn a bonus of up to $20 depending on the thoughtfulness and consistency of their answers.

The experimental manipulations play a limited role in this report, because no confirmatory hypothesis is a test of a treatment effect. The confirmatory tests of H1–H3 and H4a pool respondents across arms. The two treatments instead aim to inform the interpretation of the replication results: if low measured support reflects difficulty parsing lists of numbers, proportional visual representations should raise it; if it reflects low effort, paying for thoughtful and consistent answers should raise it. The results related to these experimental arms are reported as exploratory (Section 3.3; deviation 9 in Section 3.4). The Incentivized arm serves one further purpose: respondents in this arm could subsequently donate all, part, or none of the earned bonus to an organization working to reduce inequality, and this donation supplies the behavioral measure for H4b, which is the one registered test estimated within a single arm.

Orthogonal to the arms, a second randomization varied whether the exploratory items on inequality perception and evaluation appeared before or after the replication items. This randomization is non-focal: no hypothesis or arm comparison depends on it; it guards against order effects on the replication items, and it shows none (Section 3.3). Crossing it with the two non-incentivized arms, plus the single-version incentivized arm, produces the five questionnaire versions used throughout (Table 4). Randomization was implemented in the survey software after consent.

2.2 Sampling, recruitment, and realized sample

Respondents were recruited through the online access panel of Bilendi with quotas on gender, age, and education set to the U.S. adult population. The target sample size was 3,300 individuals. The sample size was determined by the availability of resources; it is nonetheless more than three times that of the original study (N = 1,108) and allows heterogeneity analyses.

Fieldwork ran from 05 March 2026 to 27 May 2026. Of the 7,172 respondents who entered the survey, 3,695 reached the final page: 1,631 in the Control arm, 1,723 in the Visual Cues arm, and 341 in the Incentivized arm. Median completion time was 11.6 minutes (interquartile range 7.6–17.5). Appendix A reports entries and completions by version and arm against the registered plan, together with the sociodemographic composition of both analysis samples (Table 4, Table 5).

2.3 Measures

Numeric comparison items (Q1–Q9). Respondents compared two stylized income distributions, each consisting of a few income values, and judged which was more unequal, or whether the two were similarly unequal, in the original Amiel–Cowell numeric format. The fielded items are reproduced in Table 22 in Appendix F. Q1 doubles all incomes (scale invariance); Q2 adds a fixed sum to every income (translation invariance); Q3 replicates the population (population/replication principle). Q4 tests the transfer principle: its distribution B is obtained from distribution A by a rank-preserving transfer from a richer to a poorer person, so that B Lorenz-dominates A. Q5 and Q6 test decomposability by presenting one pair of distributions and then the same pair with an identical third subgroup merged into each. Three further items (Q7–Q9) compare three overlapping distributions pairwise and support a transitivity check. The sets were randomly ordered in all questions. Every item offers the same options: checking A, checking B, or checking "A and B". Following Amiel & Cowell (1992), support for scale invariance, translation invariance, and the population principle is coded as judging the two distributions similarly unequal. Support for the transfer principle is coded as judging the Lorenz-dominated distribution more unequal. The two remaining responses correspond to Amiel and Cowell's "strongly disagree" (the Lorenz-dominating distribution judged more unequal) and "disagree" (the two judged equivalent). Support for decomposability is coded as giving the same ranking in Q5 and Q6, since the two questions differ only by adding an identical subgroup to both distributions.

Verbal axiom items (Q10–Q14). Each axiom was also stated in words, with respondents choosing among substantive interpretations. Q10 covers scale invariance, Q11 translation invariance (in an "add a fixed sum" and a "deduct a fixed sum" variant), Q12 the population principle, Q13 the transfer principle, and Q14 decomposability. Support is coded as selecting the option that states the axiom. Table 23 in Appendix F shows all items.

Redistribution preferences. Two items asked whether the government (public actor) and major companies (private actor) ought to reduce income or pay differences with 7-point response scales (Lindh & McCall, 2023). Higher values indicate stronger support for redistribution.

Leaky bucket. Three items adapted from Amiel, Creedy, and Hurn (1999) asked, for $100, $1,000, and $10,000 taken from richer people, the minimum amount that must reach poorer people for the respondent to support the transfer, with an explicit "Don't know" option. The registered index is the mean *tolerated proportional loss* across the three items, where each item's tolerated loss is 1 − (required amount / amount taken). Higher values thus denote greater willingness to sacrifice efficiency for equality. The implemented index is the mean of the one to three valid item responses available for a respondent.

Inequality-reduction behavior. In the incentivized arm, respondents who earned a bonus could donate all, part, or none of it to an organization described as "a non-profit organization working to reduce inequality". The primary measure is the binary indicator of donating any amount.

Controls. Age group, gender, education, and left–right political self-placement.

Item non-response. The axiom items have a high response rate (item-level n is reported in Table 9 in Appendix A). The exception is the leaky-bucket block: only 55.0% of completers supplied a usable value on at least one of the three items, the rest almost always selecting the explicit "Don't know" option, a volume not anticipated (deviation 6 in Section 3.4). Respondents with no valid leaky-bucket item are missing on the H4a index, so the H4a estimate is conditional on willingness to attempt the task (Section 4.5).

2.4 Analytical samples

We report every confirmatory result on two samples. The analytical sample (N = 3,695) comprises everyone who reached the final questionnaire page. The robustness sample (N = 2,125) applies the registered data-quality screens: it retains completers who passed two attention checks and spent at least one minute on the numeric-inequality block. The registered plan treated the screened set as the analysis population. The screens, however, remove 42.5% of completers, and discarding that large a share of a quota sample would be a consequential analytic choice, especially for group-specific analyses. We therefore treat all completers as the analytical sample and the screened set as the robustness sample, documented in Section 3.4. Table 6 in Appendix A reports the exclusion funnel: the completers excluded by each data-quality criterion and by their union, which defines the robustness sample. Section 3.2 shows that the two samples agree on all confirmatory results.

Per-covariate balance diagnostics across the five questionnaire conditions show no imbalance in either sample that would challenge the treatment comparisons (Table 8 in Appendix A).

2.5 Analysis plan

The registered analysis plan specifies one test procedure per hypothesis. All are implemented as registered except where Section 3.4 states otherwise.

- **H1**: one-sample proportion tests of transfer-principle support against H_0: $\pi \geq 0.66$, one-tailed, run separately for the numeric and the verbal item.
- **H2**: a logistic generalized linear mixed model with a respondent random intercept and axiom type as a within-subject fixed factor, with the transfer principle as reference. The registered contrasts are the four other axioms against the transfer principle.
- **H3**: linear regressions of (a) public and (b) private redistribution preference on binary numeric transfer-principle support (Q4).
- **H4**: (a) a linear regression of the mean tolerated proportional loss on binary numeric transfer-principle support, with a registered item-level robustness model using respondent random intercepts; and (b) a logistic regression of donating any of the bonus on transfer-principle support, in the incentivized arm only.

Tests are two-tailed at $\alpha = .05$ except for H1. Multiplicity is controlled with Holm's step-down procedure within each hypothesis family (two tests for H1, four contrasts for H2, two for H3, two for H4). Registered secondary analyses re-estimate the H2–H4 models with controls for gender, age group, education, and left–right political orientation. For H3 and H4, we focus on the numeric format (Q4) of the transfer principle.

2.6 Ethics and consent

The DIW Berlin Ethics Committee approved the project on October 22, 2025, before data collection, and all respondents gave informed consent.

3 Results

3.1 Descriptive results

Figure 1 shows the proportion of respondents whose answers are consistent with each axiom, in both question formats and both analysis samples, against the 0.66 threshold used in H1 (see Table 9 in Appendix A for complete results). Apart from the registered transfer-principle tests in H1, the verbal-item estimates and numeric–verbal comparisons in this subsection are descriptive.

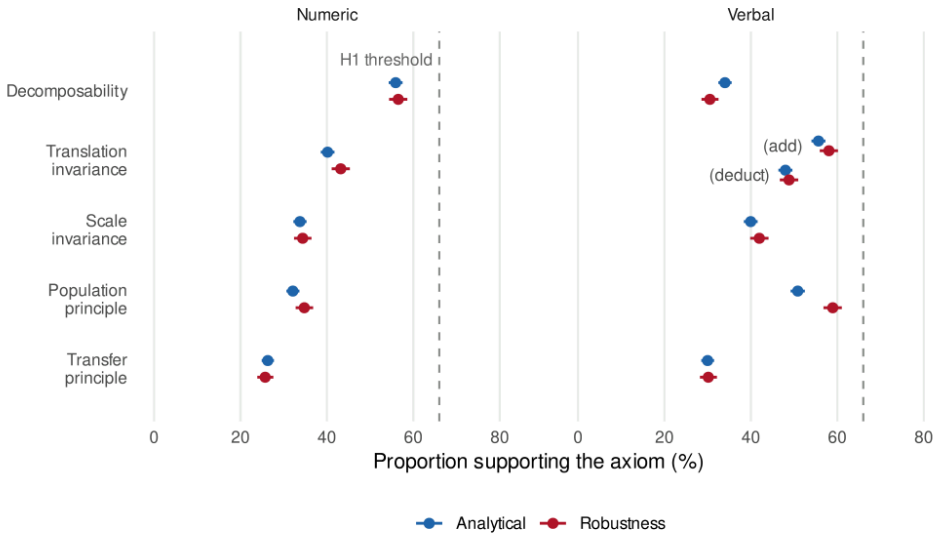


Figure 1. Support for each inequality axiom by question format (panels) and analysis sample (color). Points are proportions with 95% confidence intervals; both panels share the same rows, ordered by numeric-format support in the analytical sample. The dashed line marks the 66% threshold used in H1. In the verbal panel, the translation-invariance row carries the 'add a fixed sum' and 'deduct a fixed sum' variants of Q11 as separate annotated point pairs.

Two patterns of results stand out compared to the published percentages of the original study (Table 19 in Appendix C). First, support for the transfer principle in the general-population sample is lower than in the original student samples, in both formats. Second, the format effect that Amiel and Cowell documented, where verbal statements attracted substantially more agreement than numerical comparisons, does not reproduce for the transfer principle: numeric and verbal support are close here (26.3% against 30.0%), whereas Amiel and Cowell found 35% against 60%.

Descriptive statistics for the extension outcomes are in Table 10 in Appendix A. Missingness in the leaky-bucket block leaves 55.0% of respondents with an H4a index value. In the incentivized arm, the mean earned bonus was $17.88, the mean amount donated was $2.91, and 27.8% of respondents with a donation decision donated at least something. Both the missingness and the donation figures matter for interpreting H4; we return to them in Section 3.4 and Section 4.5.

3.2 Confirmatory results

Table 1 relates each hypothesis to its registered test, the observed outcome, and the registered prospective interpretation quoted from the Study Design Table. Results in the table are shown for the analytical sample, while robustness-sample conclusions are identical throughout (Table 2).

Table 1. Confirmatory results summary: test, analytical-sample outcome, and prospective interpretation per hypothesis.

Test	Outcome	Conclusion	Prospective interpretation
H1: Transfer-principle support below 0.66			
One-tailed proportion test vs. $\pi \geq 0.66$, per format (numeric, verbal)	Numeric 26.3%, verbal 30.0%; both far below 0.66 (largest Holm-adjusted p < .001)	Supported	"If the proportion … is significantly greater than or equal to 0.66, this will be interpreted as limited or very limited disagreement … Otherwise, it will confirm that support for the transfer principle is contentious in a general population sample."
H2: Transfer principle least supported of five axioms (numeric)			
Logistic GLMM; four contrasts vs. the transfer principle	All four contrasts positive; OR 1.37–4.15, largest Holm-adjusted p < .001	Supported	"If support for the other axioms is consistently and significantly higher than for the transfer principle, this replicates the claim that the transfer principle is relatively less supported among core axioms."
H3a: Support predicts public-redistribution preference			
Linear regression on binary transfer support (Q4)	b = 0.107, 95% CI [−0.043, 0.257], p = .163	Not supported	"If there is no association … this suggests that agreeing with the transfer principle may reflect task-specific reasoning or comprehension rather than general pro-redistribution attitudes."
H3b: Support predicts private-redistribution preference			
Linear regression on binary transfer support (Q4)	b = 0.039, 95% CI [−0.104, 0.183], p = .590	Not supported	Same as H3a.
H4a: Support predicts tolerated efficiency loss			
Linear regression on transfer support; item-level mixed model as robustness	b = 0.014, 95% CI [−0.018, 0.046], p = .384	Not supported	"If supporters of the transfer principle accept systematically larger efficiency losses, this supports the interpretation that endorsement of the principle reflects inequality aversion. If there is no relationship … this challenges the idea that agreeing with the transfer principle is a good attitudinal marker of inequality aversion."
H4b: Support predicts donating the earned bonus			
Logistic regression on transfer support, incentivized arm only	OR = 1.27, 95% CI [0.77, 2.10], p = .344	Not supported	"If there is no relationship … this challenges the idea that agreeing with the transfer principle is a good behavioral marker of inequality aversion."

Complete test specifications in the Analysis plan (Data and Methods); all tests are two-tailed at $\alpha = .05$ except H1 (one-tailed). Outcomes from the analytical sample; OR = odds ratio; p-values to three decimals, with values below .001 shown as < .001. Interpretations are quoted verbatim from the Study Design Table. Detailed per-test tables, with exact p-values, are in Appendix B.

Table 2 provides a comprehensive overview of all ten registered tests for both analysis samples. Below, we describe the results for each hypothesis (see Appendix B for detailed results).

Table 2. Tests in both analysis samples.

Test	n	Estimate	95% CI	Statistic	p-value	p (Holm)	Conclusion
Analytical sample							
H1: numeric (Q4)	3,687	26.3%***	[24.9%, 27.8%]	$\chi^2 = 2588.5$	< .001	< .001	Supported
H1: verbal (Q13)	3,623	30.0%***	[28.5%, 31.5%]	$\chi^2 = 2095.3$	< .001	< .001	Supported
H2: scale invariance	3,689	OR = 1.48***	[1.34, 1.64]	z = 7.48	< .001	< .001	Supported
H2: translation invariance	3,689	OR = 2.02***	[1.83, 2.24]	z = 13.59	< .001	< .001	Supported

Test	n	Estimate	95% CI	Statistic	p-value	p (Holm)	Conclusion
H2: population principle	3,689	OR = 1.37***	[1.23, 1.52]	z = 5.88	< .001	< .001	Supported
H2: decomposability	3,689	OR = 4.15***	[3.75, 4.59]	z = 27.61	< .001	< .001	Supported
H3a: public redistrib.	3,663	b = 0.107	[−0.043, 0.257]	t = 1.40	.163	.325	Not supported
H3b: private redistrib.	3,662	b = 0.039	[−0.104, 0.183]	t = 0.54	.590	.590	Not supported
H4a: tolerated loss	2,029	b = 0.014	[−0.018, 0.046]	t = 0.87	.384	.688	Not supported
H4b: donated any bonus	337	OR = 1.27	[0.77, 2.10]	z = 0.95	.344	.688	Not supported
Robustness sample							
H1: numeric (Q4)	2,124	25.7%***	[23.9%, 27.6%]	χ^2 = 1536.8	< .001	< .001	Supported
H1: verbal (Q13)	2,094	30.1%***	[28.2%, 32.1%]	χ^2 = 1200.4	< .001	< .001	Supported
H2: scale invariance	2,124	OR = 1.58***	[1.38, 1.81]	z = 6.58	< .001	< .001	Supported
H2: translation invariance	2,124	OR = 2.39***	[2.09, 2.73]	z = 12.80	< .001	< .001	Supported
H2: population principle	2,124	OR = 1.61***	[1.40, 1.84]	z = 6.85	< .001	< .001	Supported
H2: decomposability	2,124	OR = 4.35***	[3.80, 4.97]	z = 21.60	< .001	< .001	Supported
H3a: public redistrib.	2,120	b = 0.151	[−0.052, 0.355]	t = 1.46	.145	.290	Not supported
H3b: private redistrib.	2,118	b = 0.073	[−0.117, 0.262]	t = 0.75	.452	.452	Not supported
H4a: tolerated loss	1,274	b = 0.014	[−0.025, 0.054]	t = 0.72	.473	.641	Not supported
H4b: donated any bonus	220	OR = 1.38	[0.73, 2.61]	z = 0.99	.321	.641	Not supported

Estimates are the proportion supporting the transfer principle (H1), odds ratios (OR) against the transfer principle (H2) or of donating (H4b), and differences between supporters and non-supporters of the transfer principle (H3, H4a). H1 p-values are one-tailed as registered; all others two-tailed. Holm adjustment is applied within each hypothesis family and sample. p-values are reported to three decimals, with values below .001 shown as < .001; exact values are in Appendix B. Stars mark Holm-adjusted significance at the .05 (one star), .01 (two stars), and .001 (three stars) levels.

3.2.1 H1: Support for the transfer principle

H1 is supported, by a wide margin. The observed proportion is considerably below the registered two-thirds threshold, being close to one-quarter in numeric form. Thus, the registered alternative reading of "limited or very limited disagreement" (Table 1) is not justified.

In the analytical sample, 26.3% of respondents endorsed the transfer principle in the numeric format, far below the registered threshold (95% CI [24.9%, 27.8%], $\chi^2(1) = 2588.5$, Holm-adjusted one-tailed p = 1.3111e-564, Cohen's $h = -0.82$). The verbal item gives a slightly higher support rate with the same conclusion: 30.0% (95% CI [28.5%, 31.5%], $\chi^2(1) = 2095.3$, Holm-adjusted p = 8.8691e-458). The robustness sample gives 25.7% (numeric) and 30.1% (verbal), with the same conclusion. Under the joint criterion described in the narrative plan—endorsing the principle in both formats (see Section 3.4)— only 9.7% of respondents with both items qualify as supporters and 53.5% reject it in both (n = 3,618).

The three-way Amiel–Cowell classification of the numeric item shows how respondents departed from the principle: in the analytical sample 26.3% agreed, 33.1% judged the post-transfer distribution *more* unequal ("strongly disagree"), and 40.6% judged the two distributions similarly unequal ("disagree"). The corresponding split in the original study was 35% / 42% / 22%.

3.2.2 H2: Relative support for the transfer principle among the five axioms

H2 is supported. The transfer principle is the least supported of the five axioms in numeric form, and the ordering is unambiguous: no other axiom's confidence interval approaches parity with it (Figure 2).

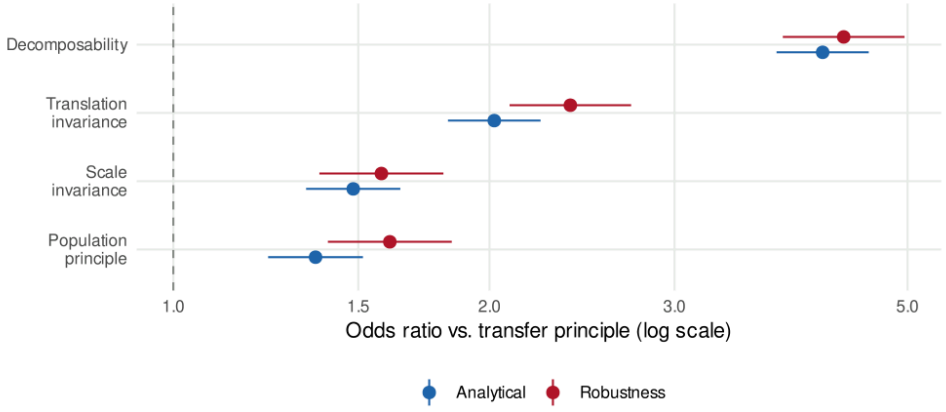


Figure 2. H2. Every other axiom attracts more support than the transfer principle. Odds of supporting each axiom relative to the transfer principle, from the mixed-effects logistic model with respondent random intercepts, by analysis sample. Points are odds ratios with 95% confidence intervals on a logarithmic scale; the dashed line marks parity with the transfer principle.

Every registered contrast is positive and significant after Holm adjustment in both samples (largest Holm-adjusted p in the analytical sample = 4.154e-09). In that sample, each of the other four axioms attracts higher odds of support than the transfer principle: decomposability (OR = 4.15, 95% CI [3.75, 4.59]), translation invariance (OR = 2.02, 95% CI [1.83, 2.24]), the population principle (OR = 1.37, 95% CI [1.23, 1.52]), and scale invariance (OR = 1.48, 95% CI [1.34, 1.64]).

3.2.3 H3: Transfer-principle support and redistribution preferences

Neither H3a nor H3b is supported. Considering the null findings, the confidence intervals are informative: on a 7-point scale, positive differences larger than about 0.26 scale points are excluded.

Transfer-principle supporters report slightly stronger redistribution preferences than non-supporters, but neither difference is distinguishable from zero. In the analytical sample the public-actor difference is 0.107 scale points (95% CI [−0.043, 0.257], t = 1.40, p = 0.1626, Holm-adjusted p = 0.3252, n = 3,663) and the private-actor difference 0.039 scale points (95% CI [−0.104, 0.183], t = 0.54, p = 0.5898, Holm-adjusted p = 0.5898, n = 3,662).

3.2.4 H4: Transfer-principle support and inequality aversion

Neither H4a nor H4b is supported. Unlike the H3 intervals, the H4b confidence interval is wide given the small sample size, so the test cannot rule out moderate associations in either direction.

H4 tests two outcomes. H4a is the leaky-bucket index, where higher values mean greater inequality aversion. H4b is whether respondents donated any of their earned bonus (incentivized arm only, see Table 14 and Table 15 in Appendix B for complete results, and Figure 3 for the difference estimates for each of H3 and H4a).

Both H4 estimates point in the predicted direction, and neither is statistically significant. In the analytical sample, supporters tolerate a mean proportional loss 0.014 higher than non-supporters (95% CI [−0.018, 0.046], t = 0.87, p = 0.3838, Holm-adjusted p = 0.6883, n = 2,029). The registered item-level robustness model with respondent random intercepts agrees (Table 16): b = 0.016, 95% CI [−0.016, 0.047], z = 0.96, p = 0.3351, from 5,791 item responses by 2,029 respondents.

For donation behavior, 30.9% of supporters donated at least part of their bonus against 26.0% of non-supporters (OR = 1.27, 95% CI [0.77, 2.10], z = 0.95, p = 0.3442, Holm-adjusted p = 0.6883, n = 337).

3.2.5 Covariate-adjusted analyses

The preregistration specifies, as registered secondary analyses, re-estimating the H2–H4 models with controls for gender, age group, education, and left–right political self-placement. After these adjustments, the results are equivalent to the unadjusted analyses. All four H2 contrasts remain positive and significant

after Holm correction, and no H3 or H4 coefficient reaches significance in either sample (Table 17 and Table 18 in Appendix B).

3.2.6 Summary of confirmatory results

Of the ten confirmatory tests, 6 met the registered $\alpha = .05$ criterion in the predicted direction in the analytical sample: both H1 tests and all four H2 contrasts (Table 2). The four extension tests (H3a, H3b, H4a, H4b) did not. Conclusions are identical in the robustness sample. As a sensitivity check, applying Holm across all ten tests jointly rather than within families leads to the same conclusions.

3.3 Exploratory analyses

The arm comparisons were considered in the research plan but not registered as confirmatory tests with decision rules (see Section 3.4). We summarize them here because they inform the confirmatory outcomes. The supporting tables and figures are in Appendix E. The exploratory items on inequality perception and evaluation are not analyzed in this report.

3.3.1 Experimental arms

The two manipulations behaved differently (Table 20 and Figure 4 in Appendix E). Visual cues did not significantly change endorsement of the transfer principle (OR = 1.13 relative to Control, 95% CI [0.96, 1.31], $p = 0.1362$). Respondents in the incentivized arm endorsed the transfer principle more often than those in Control (32.6% against 24.5%; OR = 1.49, 95% CI [1.16, 1.92], $p = 0.0019$, Holm-adjusted $p = 0.0038$), with a similar estimate in the robustness sample (OR = 1.53, 95% CI [1.11, 2.11]). The registered non-focal order randomization had no detectable effect (OR = 1.02, 95% CI [0.88, 1.18], $p = 0.8049$ in the analytical sample).

Two cautions apply to the incentive result. The incentivized arm is by far the smallest, so the comparison is estimated with limited precision. And the shift itself is modest: from roughly a quarter to roughly a third, still far below the 0.66 threshold. Thus, the H1 conclusion holds within the incentivized arm.

3.3.2 Who endorses the transfer principle?

The clearest correlate of endorsing the transfer principle is education (Table 21 in Appendix E): in the analytical sample, college-educated respondents have 1.65 times the odds of endorsement (95% CI [1.41, 1.92]). In contrast, left–right self-placement is essentially unrelated (OR = 1.00 per scale point, 95% CI [0.98, 1.03], $p = 0.8836$). The robustness sample shows the same pattern (college OR = 1.61, left–right OR = 1.00). The fact that an education gradient exists where a political gradient does not is consistent with the idea that the axiom items behave more like a comprehension task than like a measure of distributive preferences or beliefs. The gradient is modest in absolute terms, as endorsement remains a minority position in every educational group.

3.3.3 Response consistency

Among analytical-sample respondents who answered all three transitivity items (Q7–Q9), 13.1% (484 of 3,682) gave the same ranking in Q7 and Q8 but a different one in Q9, violating transitivity (13.1% in the robustness sample), close to the roughly 11% Amiel and Cowell reported. Agreement across formats is similarly weak: as reported under H1, only 9.7% of respondents endorsed the transfer principle in *both* the numeric and the verbal item, while 53.5% rejected it in both (n = 3,618).

3.4 Deviations from the preregistration

Departures from the Research Plan and Study Design Table are summarized in Table 3 (also in Appendix D for details). None of the deviations seem to impact the conclusions: all ten registered primary tests were estimable and are reported, together with the covariate-adjusted and H4a item-level checks.

Table 3. Summary of all deviations from the preregistration.

No.	Item	Registered	Implemented here
1	Realized sample	3,300 completers (1,500/1,500/300)	3,695 completers; slightly above target in every arm
2	Multiplicity procedure	Romano–Wolf step-down	Holm step-down; no conclusion could differ at $\alpha = .05$

No.	Item	Registered	Implemented here
3	Analysis sample	Screened sample as analysis population	All completers analytical; screened set as robustness
4	Multiplicity family	'Each axiom as a family' (ambiguous)	Within-hypothesis families; joint-Holm sensitivity check
5	Panel referrer	Not specified	Exact-match referrer filter (6 records removed)
6	Leaky-bucket DK volume	Not anticipated	45.0% of completers without any usable item
7	H1 verbal operationalization	Narrative: joint criterion; Design Table: per format	Per format (Design Table); joint criterion descriptive
8	H3/H4 predictor	Binary support; format unspecified	Numeric Q4, consistent with H2
9	Treatment-arm comparisons	Described without contrasts or decision rules	Reported as exploratory

Details and quantities are given in the numbered text of Appendix D.

4 Discussion

4.1 Conclusions by hypothesis

H1 replicates the original finding and strengthens it. The transfer principle is endorsed by only a minority in a U.S. general-population sample: support is 26.3% in numeric form and 30.0% in verbal form, in both cases far below the original student sample figures of 35% and 60%. We interpret this as supporting evidence that the transfer principle is contentious among laypeople. Our result shows this in a stronger form than anticipated: at roughly one respondent in four, the numeric result is a clear minority position.

H2 is likewise supported: as anticipated, the transfer principle is the least supported of the five axioms, with all four registered contrasts significant after correction and odds ratios from 1.37 to 4.15. In terms of proportions, numeric support for the other four axioms ranges from 32.1% to 55.9%, against 26.3% for the transfer principle. This sharpens the original pattern: there, translation invariance was narrowly the least supported axiom in numeric form; here the transfer principle falls significantly below every other axiom (see below). Our results show that the transfer principle is distinctively at odds with lay judgment compared to the other axioms.

H3, in contrast, does not hold: endorsing the transfer principle is not associated with stronger redistribution preferences for either public or private actors, and the confidence intervals exclude positive differences larger than about a quarter of a scale point on a 7-point measure. The absence of an association suggests that support for the transfer principle reflects task-specific reasoning or comprehension rather than general pro-redistribution attitudes. In other words, support for the transfer principle is not a good proxy for the public's broader distributive inclinations.

Nor does H4 hold: neither the leaky-bucket index nor incentivized donation behavior is significantly related to transfer-principle support, though both point weakly in the predicted direction. For H4a the estimate is precise enough to rule out more than a small association; for H4b the interval is wide and the null is uninformative about moderate effects given its small sample size. This absence of a relationship challenges the idea that agreeing with the transfer principle is a good attitudinal or behavioral marker of inequality aversion.

4.2 What the outcomes mean for the shared-foundations claim

This study aimed to test whether support for expert axioms is a good proxy for the public's broader distributive inclinations. The four hypotheses jointly give a coherent and negative answer.

The replication part clearly succeeds. The core descriptive claim of Amiel & Cowell (1992)—that laypeople do not broadly share the axiomatic foundations of inequality measurement, and that the transfer principle is the point of strongest divergence—holds in a general adult population more than three decades

later, in a different survey mode, at 3.3 times the original sample size. The low support for the transfer principle is therefore not confined to the original student samples: lay-expert divergence appears as an empirical regularity.

The extension part contributes to understanding the replication through its null findings. If rejecting the transfer principle signaled different distributive norms, supporters and non-supporters should differ on redistribution preferences and inequality aversion. They do not. One plausible interpretation is that responses to the axiom items partly track how people approach an unfamiliar comparison task—making sense of what is more unequal when facing sets of numbers—rather than normative preferences about economic inequality. The exploratory correlates are consistent with that interpretation: endorsement is predicted by education but not by political orientation. This pattern thus points to a comprehension-sensitive judgment rather than an ideological stance.

Two further features of the data point the same way: 13.1% of respondents violate transitivity across Q7–Q9, and only 9.7% endorse the transfer principle in both formats. The Visual Cues arm, however, shows no such display effect: proportional representations of the compared distributions did not move endorsement, so clearer visual displays do not appear to change the low support for the transfer principle. The one condition in which endorsement differed was the incentivized arm, where about a third of respondents endorsed the principle rather than about a quarter, which still falls well short of broad acceptance of the principle.

The shared-foundations claim therefore fails on both counts: laypeople do not broadly endorse the axioms, and endorsing them does not stand in for distributive preferences. This is a weaker conclusion than saying that the public rejects the transfer principle on normative grounds, and a different one from saying that the public simply misunderstands the question. Comprehension plausibly shapes the responses, as the education gradient suggests; what the data rule out is that comprehension is the whole story, since the divergence persists under proportional visual aids, under a bonus for careful answers, and among respondents who passed both attention checks and did not rush the items. The response pattern is reproducible, but we find no evidence that what it measures also drives distributive policy attitudes. For inequality measurement the implication holds whatever the source of the divergence: expert axioms cannot be assumed to describe how the public reasons about inequality, and endorsement of them is not a usable proxy for distributive preferences.

4.3 Comparison with the original study

Three differences from Amiel & Cowell (1992) stand out. First, support is lower here for four of the five axioms in numeric form — narrowly so for decomposability (55.9% here against 57%) — with translation invariance the exception (40.2% here against 31%). Second, the full ordering differs: in the original study, translation invariance (31%) was slightly less supported than the transfer principle (35%); here the transfer principle is significantly below translation invariance and every other axiom. Third, and most notably, the numeric-versus-verbal gap for the transfer principle does not reproduce. Amiel and Cowell reported 35% numeric against 60% verbal and read the gap as evidence that verbal argument could persuade where numerical examples failed. Here the two formats give nearly the same answer (26.3% against 30.0%). One plausible explanation is composition: their respondents were economics and social-science students, for whom the verbal statement of the principle may have been familiar in a way it is not for a general population. Amiel and Cowell themselves found that Lorenz-trained economics students answered the verbal question far more conventionally than their non-economist peers. What does reproduce closely is the transitivity check (13.1% violations here against roughly 11% there), showing equivalent response quality across both studies.

4.4 Deviations and their implications

The deviations documented in Section 3.4 are individually minor and jointly immaterial to the conclusions. The two most consequential are the substitution of Holm for Romano–Wolf (deviation 2) and the re-designation of the analysis samples (deviation 3). The first cannot change any conclusion on these data: the gap between the largest significant raw p-value (4.154e-09) and the smallest non-significant one (0.1452) is so wide that any familywise-error-controlling procedure at $\alpha = .05$ yields the same ten conclusions. The second is reported symmetrically through every test on both samples: the two samples agree throughout, so no conclusion depends on the choice of analysis sample. The remaining deviations are operational clar-

ifications of underspecified registered text, each resolved before analysis in the direction most consistent with the Study Design Table.

4.5 Limitations

We acknowledge some limitations. First, the attention-check failure rate was rather high, at 28.9% on the factual item alone. Reporting every result on both the analytical (unscreened) and the robustness (screened) sample addresses the resulting sensitivity, and the two agree throughout, but the overall quality of online panel responses to a demanding comparison task remains a real constraint. Failing an attention check is also socially patterned: completers who failed differ from those who passed in age, gender, highest degree, and political self-placement, though not in college education (Table 7 in Appendix A). The robustness sample therefore conditions on a non-random subset of completers, which also justified reporting results on both samples.

Second, about 45.0% of completers declined all three leaky-bucket items. If willingness to answer correlates with inequality aversion, the H4a estimate is conditional in a way the registered design did not anticipate. Third, the incentivized arm had 341 completers, so the behavioral test cannot distinguish a null from a moderate association.

Finally, the Amiel–Cowell items were retained verbatim for comparability, which preserves their ambiguities. The main one is the shared third response of the numeric items: checking "A and B" to say the two distributions are similarly unequal may express a considered equality judgment, but also indifference, incomparability, or a default answer under uncertainty. Under the original coding, which we follow, this ambiguity cuts both ways. For scale invariance, translation invariance, and the population principle, "A and B" is the response coded as *supporting* the axiom, so ambiguous responding can overstate support for these axioms. For the transfer principle it is coded as non-support, where it can understate support. And for decomposability, any ranking repeated across Q5 and Q6, including a repeated "A and B", counts as support.

4.6 Conclusion

Most laypeople do not endorse the transfer principle, the axiom that inequality measurement treats as most central. Roughly one U.S. adult in four accepts it in numeric form, and it is the least supported of five core axioms, thus supporting the H1 and H2 predictions and extending Amiel and Cowell's finding of low transfer-principle support beyond the student populations in which it was first documented. Endorsement, however, carries no distributive meaning that we can detect: supporters and non-supporters barely differ in redistribution preferences and inequality aversion, and the confidence intervals rule out all but small associations for the stated-preference and leaky-bucket outcomes. Agreement with the axiom therefore reads less like an egalitarian conviction and more like a response to an unfamiliar comparison task. The next step for measuring public assent to the foundations of inequality measurement is to separate comprehension of that task from the normative judgment it is meant to elicit.

Declarations

Competing interests. The authors have declared that no competing interests exist.

Funding. This study was funded by the SOEP-LEE2 project. As declared in the registered Research Plan, the funder played no role in the study design, data collection, analysis, decision to publish, or submission preparation.

Ethics approval. The project was approved prospectively by the Ethics Committee of DIW Berlin (German Institute for Economic Research), which reviewed it under the title "Measuring attitudes towards inequality" at its meeting of 19 September 2025 and deemed it ethically unproblematic. The committee's confirmation letter, signed on its behalf by the head of the committee, is dated 22 October 2025; the decision is not accompanied by a protocol number. All participants gave informed consent before randomization.

Preregistration. The Research Plan was registered before data collection on 6 December 2025 (Moya et al., 2025), OSF registration 10.17605/OSF.IO/82DJS. All deviations are documented in Section 3.4.

Data availability. A minimal anonymized dataset sufficient to reproduce every statistic in this report, together with its codebook, is publicly archived with the reproduction package in the project's OSF repository (10.17605/OSF.IO/8Z2UV) under a CC BY 4.0 license. The raw data is not shared, to protect respondent privacy.

Materials availability. The visual-cue stimuli are archived with the reproduction package in the project's OSF repository (10.17605/OSF.IO/8Z2UV); the fielded item wordings are reproduced in Appendix F.

Code availability. This report is a literate Quarto document: every statistic in its text, tables, and figures is computed by the code it contains. A public reproduction package in the form of a Docker-pinned repository that re-renders this report from the anonymized minimal dataset is archived in the project's OSF repository (10.17605/OSF.IO/8Z2UV) under an MIT license.

Generative AI and assistive tooling. Parts of the analysis code and manuscript text were drafted and revised with the assistance of a large language model (Claude, Anthropic), used as an editing and programming aid under author direction. All analyses, results, and interpretations were reviewed and verified by the authors, who take full responsibility for the content.

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Appendix A: Sample, data quality, and descriptive tables

Table 4. Realized sample by experimental arm and questionnaire version, against the registered plan.

Version	Condition	Planned n	Entered	Completed	Completion rate
V1	Incentivized	300	610	341	55.9%
V2	Control, replication items first	1,500 (arm total)	1,475	762	51.7%
V3	Visual Cues, replication items first	1,500 (arm total)	1,568	846	54.0%
V4	Control, exploratory items first	(same arm as V2)	1,568	869	55.4%
V5	Visual Cues, exploratory items first	(same arm as V3)	1,577	877	55.6%
	Arm total: Control	*1,500*		*1,631*	
	Arm total: Visual Cues	*1,500*		*1,723*	
	Arm total: Incentivized	*300*		*341*	

Entries and completions are counted among panel entrants routed through the contracted referrer.

Table 5. Sociodemographic composition of the realized sample, by analysis sample.

	Analytical sample		Robustness sample	
Level	n	%	n	%
Age group				
18-29 years	820	22.2%	393	18.5%
30-39 years	715	19.4%	359	16.9%
40-49 years	661	17.9%	358	16.8%
50-59 years	621	16.8%	369	17.4%
60-69 years	633	17.1%	453	21.3%
Other	245	6.6%	193	9.1%
Gender				
Female	1,853	50.1%	1,121	52.8%
Male	1,842	49.9%	1,004	47.2%
College educated				
No	1,987	53.8%	1,157	54.4%
Yes	1,708	46.2%	968	45.6%
Highest degree				
Associate or Junior College	449	26.3%	257	26.5%
Bachelor's	725	42.4%	419	43.3%
Master's	479	28.0%	253	26.1%
PhD	55	3.2%	39	4.0%

Percentages are within-variable and exclude missing values.

Table 6. Data-quality exclusion funnel among completed questionnaires.

Criterion	n excluded	% of completers	n remaining
Failed attention check 1 (instructed color response)	125	3.4%	3,570

Criterion	n excluded	% of completers	n remaining
Failed attention check 2 (factual item)	1,069	28.9%	2,626
Sped through the numeric block (under 60 seconds)	790	21.4%	2,905
Any criterion (union)	1,570	42.5%	2,125

The criteria are not mutually exclusive; their union defines the robustness sample.

Table 7. Sociodemographic composition of completers who passed and failed the attention checks (analytical sample).

	Passed (n = 2,510)		Failed (n = 1,185)	
Level	n	%	n	%
Age group				
18-29 years	530	21.1%	290	24.5%
30-39 years	465	18.5%	250	21.1%
40-49 years	423	16.9%	238	20.1%
50-59 years	407	16.2%	214	18.1%
60-69 years	483	19.2%	150	12.7%
Other	202	8.0%	43	3.6%
Gender				
Female	1,306	52.0%	547	46.2%
Male	1,204	48.0%	638	53.8%
College educated				
No	1,330	53.0%	657	55.4%
Yes	1,180	47.0%	528	44.6%
Highest degree				
Associate or Junior College	303	25.7%	146	27.7%
Bachelor's	517	43.8%	208	39.4%
Master's	313	26.5%	166	31.4%
PhD	47	4.0%	8	1.5%

Passed = passed both attention checks; Failed = failed at least one. Percentages are within-variable and exclude missing values. Tests comparing the two groups: age group p = 4.495e-11, gender p = 9.790e-04, college education p = 0.1733, highest degree p = 0.0065, political self-placement (overall F) p = 6.086e-05.

Table 8. Experimental balance across the five questionnaire conditions, both analysis samples.

Covariate	Levels / metric	Test	p (Analytical)	p (Robustness)
Age group	6 categories	Chi-square	0.6528	0.1654
Gender	2 categories	Chi-square	0.8007	0.9153
College education	2 categories	Chi-square	0.5117	0.1331
Highest degree	5 categories	Chi-square	0.6493	0.5851
Political self-placement (0–10)	mean (SD)	Linear regression (overall F)	0.6822	0.3005

Covariate	Levels / metric	Test	p (Analytical)	p (Robustness)

Reported p-values are from omnibus chi-square tests (categorical covariates) and from the overall F-test of a linear regression of political self-placement on questionnaire version, the regression equivalent of a one-way ANOVA. They are descriptive diagnostics of the randomization, not hypothesis tests.

Table 9. Support for each inequality axiom, by question format and analysis sample.

	Analytical sample			Robustness sample		
Axiom	Support	95% CI	n	Support	95% CI	n
Numeric						
Transfer principle	26.3%	[24.9%, 27.8%]	3,687	25.7%	[23.9%, 27.6%]	2,124
Scale invariance	33.7%	[32.2%, 35.3%]	3,686	34.4%	[32.4%, 36.4%]	2,123
Translation invariance	40.2%	[38.6%, 41.7%]	3,686	43.2%	[41.1%, 45.3%]	2,123
Population principle	32.1%	[30.6%, 33.6%]	3,687	34.8%	[32.8%, 36.8%]	2,123
Decomposability	55.9%	[54.3%, 57.5%]	3,683	56.5%	[54.4%, 58.6%]	2,122
Verbal						
Transfer principle	30.0%	[28.5%, 31.5%]	3,623	30.1%	[28.2%, 32.1%]	2,094
Scale invariance	39.9%	[38.4%, 41.6%]	3,585	41.9%	[39.8%, 44.1%]	2,072
Translation invariance (add)	55.6%	[54.0%, 57.2%]	3,635	58.0%	[55.9%, 60.1%]	2,100
Translation invariance (deduct)	48.0%	[46.4%, 49.6%]	3,664	48.8%	[46.7%, 50.9%]	2,112
Population principle	50.8%	[49.2%, 52.5%]	3,553	58.9%	[56.8%, 61.0%]	2,055
Decomposability	34.0%	[32.5%, 35.5%]	3,625	30.5%	[28.6%, 32.5%]	2,095

Support is the proportion endorsing the axiom, with its two-sided 95% confidence interval; n is the number of valid responses to the item. Decomposability (numeric) is coded as giving the same ranking in Q5 and Q6.

Table 10. Descriptive statistics for the extension outcome measures, by analysis sample.

Measure	Analytical sample	Robustness sample
Public redistribution preference (1–7)	4.45 (2.04), n = 3,670	4.59 (2.09), n = 2,121
Private redistribution preference (1–7)	4.40 (1.95), n = 3,669	4.61 (1.94), n = 2,119
Tolerated proportional loss (leaky bucket, 0–1)	0.418 (0.333), n = 2,032	0.365 (0.323), n = 1,274
Donated any bonus (incentivized arm)	27.8% donated, n = 338	25.0% donated, n = 220

Entries are mean (SD) unless otherwise noted; the donation row covers the incentivized arm only.

Appendix B: Detailed confirmatory tables

Table 11. H1: one-sample proportion tests of transfer-principle support against the registered threshold of 0.66.

Format	n	Support	Prop	95% CI	Chi-sq.	p (1-tailed)	p (Holm)	h
Analytical sample								
Numeric (Q4)	3,687	970	26.3%	[24.9%, 27.8%]	2588.5	6.5554e-565	1.3111e-564	−0.82
Verbal (Q13)	3,623	1,086	30.0%	[28.5%, 31.5%]	2095.3	8.8691e-458	8.8691e-458	−0.74
Robustness sample								

Format	n	Support	Prop	95% CI	Chi-sq.	p (1-tailed)	p (Holm)	h
Numeric (Q4)	2,124	546	25.7%	[23.9%, 27.6%]	1536.8	2.0081e-336	4.0162e-336	−0.83
Verbal (Q13)	2,094	631	30.1%	[28.2%, 32.1%]	1200.4	2.4944e-263	2.4944e-263	−0.73

Tests are one-tailed, as registered. Confidence intervals are two-sided 95% intervals for the proportion; Holm adjustment is applied across the two tests within each sample; h is Cohen's effect size for the difference from 0.66.

Table 12. H2: logistic generalized linear mixed model of axiom support with a respondent random intercept.

Axiom vs. transfer	n resp.	n obs.	OR	95% CI (OR)	z	p-value	p (Holm)
Analytical sample							
Scale invariance	3,689	18,429	1.48	[1.34, 1.64]	7.48	7.239e-14	1.448e-13
Translation invariance	3,689	18,429	2.02	[1.83, 2.24]	13.59	4.802e-42	1.441e-41
Population principle	3,689	18,429	1.37	[1.23, 1.52]	5.88	4.154e-09	4.154e-09
Decomposability	3,689	18,429	4.15	[3.75, 4.59]	27.61	8.364e-168	3.346e-167
Robustness sample							
Scale invariance	2,124	10,615	1.58	[1.38, 1.81]	6.58	4.831e-11	4.831e-11
Translation invariance	2,124	10,615	2.39	[2.09, 2.73]	12.80	1.604e-37	4.811e-37
Population principle	2,124	10,615	1.61	[1.40, 1.84]	6.85	7.351e-12	1.470e-11
Decomposability	2,124	10,615	4.35	[3.80, 4.97]	21.60	1.970e-103	7.882e-103

The transfer principle is the reference category; odds ratios above 1 indicate that the axiom is more widely supported than the transfer principle. Holm adjustment is applied across the four registered contrasts within each sample.

Table 13. H3: linear regressions of redistribution preference on binary numeric transfer-principle support.

Outcome	n	Mean, non-supp.	Mean, supp.	b (diff.)	95% CI	t	p-value	p (Holm)
Analytical sample								
H3a: Public	3,663	4.42	4.52	0.107	[−0.043, 0.257]	1.40	0.1626	0.3252
H3b: Private	3,662	4.39	4.43	0.039	[−0.104, 0.183]	0.54	0.5898	0.5898
Robustness sample								
H3a: Public	2,120	4.55	4.70	0.151	[−0.052, 0.355]	1.46	0.1452	0.2905
H3b: Private	2,118	4.59	4.66	0.073	[−0.117, 0.262]	0.75	0.4522	0.4522

Redistribution preference runs 1–7, higher = stronger support for redistribution. Coefficients are the difference in mean preference between supporters and non-supporters. Holm adjustment is applied across the two tests within each sample.

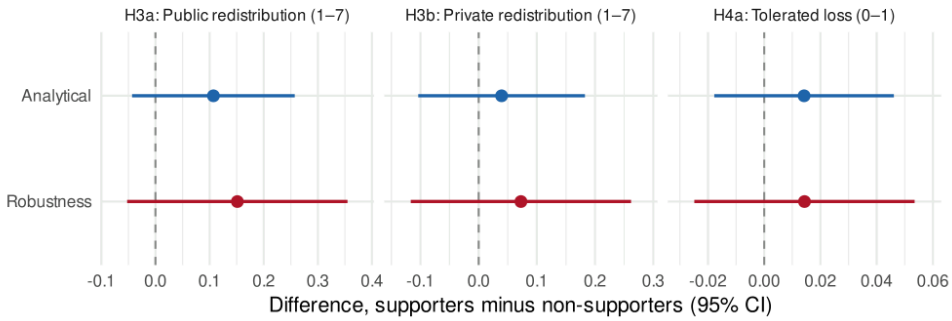


Figure 3. H3 and H4a. Differences between transfer-principle supporters and non-supporters with 95% confidence intervals, by analysis sample. Units and axis ranges differ by panel — scale points (1–7) for the two redistribution preferences, proportion of tolerated loss (0–1) for the leaky-bucket index — so magnitudes are not visually comparable across panels. The dashed line marks no difference. H4b is reported in its own metric in Table 15.

Table 14. H4a: linear regression of the mean tolerated proportional loss on binary numeric transfer-principle support.

Sample	n	Mean, non-supp.	Mean, supp.	b	95% CI	t	p-value	p (Holm)
Analytical	2,029	0.414	0.428	0.014	[−0.018, 0.046]	0.87	0.3838	0.6883
Robustness	1,274	0.361	0.375	0.014	[−0.025, 0.054]	0.72	0.4732	0.6410

The outcome runs 0–1, higher = more inequality averse. Holm adjustment is applied across the two H4 tests within each sample.

Table 15. H4b: logistic regression of donating any of the earned bonus on binary numeric transfer-principle support.

Sample	n	% donating, non-supp.	% donating, supp.	OR	95% CI (OR)	z	p-value	p (Holm)
Analytical	337	26.0%	30.9%	1.27	[0.77, 2.10]	0.95	0.3442	0.6883
Robustness	220	23.0%	29.2%	1.38	[0.73, 2.61]	0.99	0.3205	0.6410

Incentivized arm only; the estimate is an odds ratio. Holm adjustment is applied across the two H4 tests within each sample.

Table 16. H4a robustness: registered linear mixed model of item-level tolerated proportional loss (0–1) on transfer-principle support.

Sample	n respondents	n item responses	b	95% CI	z	p-value
Analytical	2,029	5,791	0.016	[−0.016, 0.047]	0.96	0.3351
Robustness	1,274	3,674	0.016	[−0.023, 0.055]	0.79	0.4273

Linear mixed models regress tolerated proportional loss on transfer-principle support with respondent random intercepts. The p-value is the two-sided Wald test for the transfer-support coefficient; this re-expression of H4a is not counted as an additional member of the H4 multiplicity family.

Table 17. Registered covariate-adjusted H2 mixed model.

Axiom vs. transfer	n resp.	n obs.	OR	95% CI (OR)	z	p-value	p (Holm)
Analytical sample							
Scale invariance	3,679	18,379	1.49	[1.34, 1.65]	7.52	5.665e-14	1.133e-13
Translation invariance	3,679	18,379	2.02	[1.83, 2.24]	13.57	6.314e-42	1.894e-41
Population principle	3,679	18,379	1.36	[1.23, 1.51]	5.85	4.781e-09	4.781e-09
Decomposability	3,679	18,379	4.15	[3.75, 4.59]	27.54	5.148e-167	2.059e-166
Robustness sample							
Scale invariance	2,122	10,605	1.58	[1.38, 1.81]	6.57	4.959e-11	4.959e-11
Translation invariance	2,122	10,605	2.38	[2.09, 2.72]	12.76	2.795e-37	8.386e-37
Population principle	2,122	10,605	1.61	[1.40, 1.84]	6.85	7.565e-12	1.513e-11
Decomposability	2,122	10,605	4.35	[3.81, 4.97]	21.57	3.280e-103	1.312e-102

Controls are gender, age group, education, and left–right political self-placement. The transfer principle is the reference axiom; Holm adjustment is across the four contrasts within each sample.

Table 18. Registered covariate-adjusted models for H3 and H4.

Outcome	n	Adjusted estimate	95% CI	Statistic	p-value	p (Holm)
Analytical sample						
H3a: Public redistribution	3,657	0.114	[−0.028, 0.257]	t = 1.57	0.1159	0.2318
H3b: Private redistribution	3,657	0.061	[−0.077, 0.200]	t = 0.87	0.3861	0.3861
H4a: Tolerated proportional loss	2,028	0.006	[−0.026, 0.038]	t = 0.37	0.7131	1.0000
H4b: Donated any bonus	337	OR 1.13	[0.64, 1.96]	z = 0.42	0.6776	1.0000
Robustness sample						
H3a: Public redistribution	2,118	0.152	[−0.037, 0.341]	t = 1.58	0.1147	0.2293
H3b: Private redistribution	2,116	0.087	[−0.095, 0.270]	t = 0.94	0.3463	0.3463
H4a: Tolerated proportional loss	1,274	0.009	[−0.031, 0.048]	t = 0.44	0.6631	0.6631
H4b: Donated any bonus	220	OR 1.42	[0.71, 2.84]	z = 0.99	0.3234	0.6469

Controls are gender, age group, education, and left–right political self-placement. Only the transfer-principle coefficient is shown; H4b is reported as an odds ratio. Holm adjustment is applied within the H3 and H4 families in each sample.

Appendix C: Comparison with Amiel and Cowell (1992)

Table 19. Axiom support in this study against the published percentages of Amiel and Cowell (1992).

Axiom	Amiel and Cowell 1992	Analytical sample	Robustness sample	Difference (Analytical)
Numeric				
Transfer principle	35%	26.3%	25.7%	−8.7 pp
Scale invariance	51%	33.7%	34.4%	−17.3 pp
Translation invariance	31%	40.2%	43.2%	+9.2 pp

Axiom	Amiel and Cowell 1992	Analytical sample	Robustness sample	Difference (Analytical)
Population principle	58%	32.1%	34.8%	−25.9 pp
Decomposability	57%	55.9%	56.5%	−1.1 pp
Verbal				
Transfer principle	60%	30.0%	30.1%	−30.0 pp
Scale invariance	47%	39.9%	41.9%	−7.1 pp
Translation invariance (add)	35%	55.6%	58.0%	+20.6 pp
Translation invariance (deduct)	28%	48.0%	48.8%	+20.0 pp
Population principle	66%	50.8%	58.9%	−15.2 pp
Decomposability	40%	34.0%	30.5%	−6.0 pp

Amiel and Cowell (1992; N = 1,108 students) report percentages of the whole sample rather than of respondents to each item, so their figures are marginally conservative. The difference column compares the analytical sample with the original study.

Appendix D: Deviations in detail

The deviations summarized in Table 3 are documented here in detail.

1. Realized sample size and arm composition. The plan targeted roughly 3,300 completed questionnaires with arms of about 1,500 (Control), 1,500 (Visual Cues), and 300 (Incentivized). The realized sample is 3,695 completers (1,631 Control, 1,723 Visual Cues, and 341 Incentivized) which slightly exceeds the target overall and in every arm. No hypothesis was affected by under-recruitment.

2. Multiple-testing correction. The plan specified Romano–Wolf step-down adjusted *p*-values; we use Holm's step-down procedure (the readily reproducible `p.adjust(..., method = "holm")` routine rather than a resampling scheme). Both control the familywise error rate; Romano–Wolf can be more powerful because it exploits the joint distribution of the test statistics. The substantive consequences here are negligible: every significant test has a raw p-value at or below 4.154e-09, and every non-significant test a raw p-value at or above 0.1452, so no familywise-error-controlling procedure at $\alpha = .05$ could change a conclusion.

3. Definition of the analysis sample. The plan stated that tests would use observations remaining after deleting respondents who failed attention checks or answered the replication items in under a minute. Those screens remove 42.5% of completers (Table 6), far more than anticipated, driven mainly by the factual attention check (1,069 failures). Because discarding two fifths of a quota sample on post-treatment screens is itself a substantive analytic choice, we label all completers as the analytical sample and the exclusion-screened set as the robustness sample, reversing the ordering implied by the plan. Every confirmatory test is reported for both, and the two agree on all ten results.

4. Family definition for multiplicity control. The plan specifies adjustment "considering each axiom as a family of tests", which does not map uniquely onto the four hypotheses. Holm is applied within each hypothesis family (two tests for H1, four contrasts for H2, two for H3, two for H4). A sensitivity analysis applying Holm across all ten confirmatory tests jointly reproduces every conclusion. Relatedly, the narrative plan's H2 analysis sentence lists scale invariance, translation invariance, and replication invariance but omits decomposability; the hypothesis itself and the Study Design Table include decomposability, and the implemented model does too.

5. Exact-match restriction on the panel referrer. 6 records in the export did not exactly match the canonical contracted-panel referrer URL and were removed before any analysis (0.08% of records).

6. Leaky-bucket missingness. The plan did not anticipate the volume of "Don't know" responses to the leaky-bucket items: 45.0% of completers gave no usable value on any of the three items, reducing

H4a to 2,029 analytical-sample respondents after also requiring valid transfer-principle support. The index averages the one to three valid items available; respondents with none are missing, not imputed, so the H4a estimate is conditional on willingness to attempt at least one item.

7. Verbal-item operationalization of H1. The narrative plan describes classifying a respondent as a supporter only if they endorse the principle "consistently … separately for numeric and verbal formulations". The Study Design Table, the authoritative specification of the confirmatory tests, instead directs that the proportion be computed and tested "for each question version (numeric, verbal)". We follow the Study Design Table and test the two formats separately. The joint criterion is reported descriptively in Section 3.3, where it yields a supporter share of 9.7%—lower still, so it could not alter the H1 conclusion.

8. Transfer-support predictor for H3 and H4. The registered plans specify a binary transfer-support predictor but do not choose between the numeric item, the verbal item, and a joint indicator. We use numeric Q4, the format explicitly specified for H2, throughout the confirmatory H3 and H4 models; cross-format consistency is reported descriptively in Section 3.3.

9. Classification of the treatment-arm comparisons. The narrative plan says the effects of the two experimental treatments on the four hypotheses "will be assessed by comparing among three groups separately", but neither the plan nor the Study Design Table specifies contrasts, interaction models, multiplicity families, or decision rules for these comparisons. They are therefore reported as exploratory in Section 3.3 and carry no confirmatory weight.

Appendix E: Exploratory analyses

Table 20. Exploratory: numeric transfer-principle support by experimental arm, both analysis samples.

Sample	Arm	n	Support	95% CI	OR vs. Control	95% CI (OR)	p-value	p (Holm)
Analytical	Control	1,628	24.5%	[22.5%, 26.7%]	(reference)	–	–	–
Analytical	Incentivized	340	32.6%	[27.9%, 37.8%]	1.49	[1.16, 1.92]	0.0019	0.0038
Analytical	Visual Cues	1,719	26.8%	[24.7%, 28.9%]	1.13	[0.96, 1.31]	0.1362	0.1362
Robustness	Control	863	24.0%	[21.3%, 26.9%]	(reference)	–	–	–
Robustness	Incentivized	221	32.6%	[26.7%, 39.0%]	1.53	[1.11, 2.11]	0.0094	0.0189
Robustness	Visual Cues	1,040	25.7%	[23.1%, 28.4%]	1.09	[0.89, 1.35]	0.3970	0.3970

Odds ratios are logistic-regression contrasts against the Original Questionnaire Control arm. Holm adjustment is applied across the two contrasts within each sample.

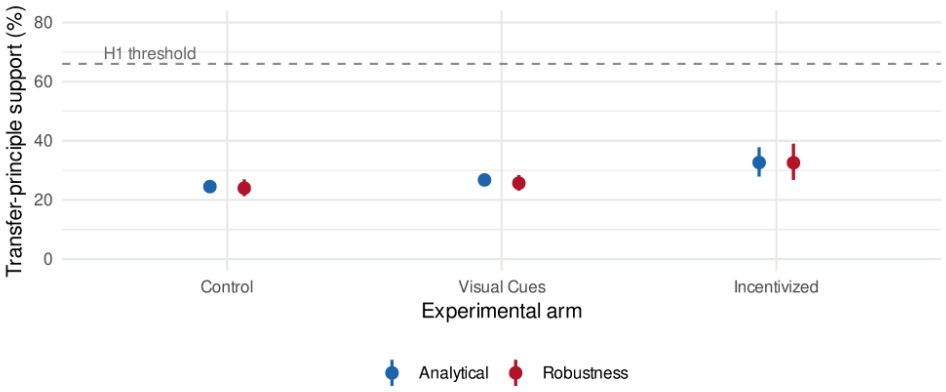


Figure 4. Exploratory. Numeric transfer-principle support by experimental arm and analysis sample. Points are proportions with 95% confidence intervals; the dashed line marks the 66% threshold from H1.

Table 21. Exploratory: logistic regressions of numeric transfer-principle support on respondent characteristics.

Predictor	n	Odds ratio	95% CI	z	p-value
Analytical sample					
College education: yes	3,677	1.65	[1.41, 1.92]	6.38	1.785e-10
Left–right placement (0–10, per point)	3,677	1.00	[0.98, 1.03]	0.15	0.8836
Gender: Male	3,677	0.99	[0.85, 1.15]	−0.13	0.9003
Age group: 30-39 years	3,677	0.88	[0.70, 1.10]	−1.14	0.2556
Age group: 40-49 years	3,677	0.79	[0.62, 1.00]	−1.98	0.0477
Age group: 50-59 years	3,677	0.89	[0.70, 1.12]	−1.00	0.3178
Age group: 60-69 years	3,677	0.61	[0.48, 0.79]	−3.83	1.257e-04
Age group: Other	3,677	0.79	[0.57, 1.10]	−1.39	0.1652
Robustness sample					
College education: yes	2,122	1.61	[1.31, 1.97]	4.60	4.301e-06
Left–right placement (0–10, per point)	2,122	1.00	[0.97, 1.04]	0.11	0.9115
Gender: Male	2,122	1.08	[0.88, 1.33]	0.77	0.4432
Age group: 30-39 years	2,122	0.96	[0.69, 1.32]	−0.27	0.7841
Age group: 40-49 years	2,122	0.80	[0.57, 1.10]	−1.37	0.1701
Age group: 50-59 years	2,122	0.82	[0.59, 1.13]	−1.23	0.2195
Age group: 60-69 years	2,122	0.65	[0.47, 0.89]	−2.67	0.0076
Age group: Other	2,122	0.84	[0.56, 1.24]	−0.88	0.3792

Reference categories are no college education, female, and the youngest age group; left–right self-placement (0 = left, 10 = right) enters linearly. Each model uses the sample's complete cases on the outcome and all four predictors.

Appendix F: Replicated questionnaire items

The replication items reproduce the Amiel & Cowell (1992) questionnaire. Every numeric item asked: "Which distribution is more unequal? If you consider that both of the distributions have the same inequality, then check 'A and B'." In the Visual Cues arm the numeric items additionally displayed proportional graphical representations of each distribution; the wording was identical. Wordings are shipped with the reproduction package (`data/minimal/items_numeric.csv` and `data/minimal/items_verbal.csv`), extracted from the survey programming and coding files (see Appendix G).

Table 22. Numeric comparison items (Q1–Q9): compared distributions and support coding.

Item	Distribution A	Distribution B	Tests	Response coded as support
Q1	(5, 8, 10)	(10, 16, 20)	Scale invariance	A and B
Q2	(5, 8, 10)	(10, 13, 15)	Translation invariance	A and B
Q3	(5, 8, 10)	(5, 5, 8, 8, 10, 10)	Population principle	A and B
Q4	(1, 4, 7, 10, 13)	(1, 5, 6, 10, 13)	Transfer principle	A
Q5	(4, 8, 9)	(5, 6, 10)	Decomposability (with Q6)	Same ranking as in Q6
Q6	(4, 7, 7, 8, 9)	(5, 6, 7, 7, 10)	Decomposability (with Q5)	Same ranking as in Q5
Q7	(5, 5, 5, 10)	(5, 5, 10, 10)	Transitivity check	–
Q8	(5, 5, 10, 10)	(5, 10, 10, 10)	Transitivity check	–
Q9	(5, 5, 5, 10)	(5, 10, 10, 10)	Transitivity check	–

Distributions as fielded. 'A and B' means checking both, i.e. judging the two distributions similarly unequal. Q7–Q9 have no support coding; they enter only the transitivity check.

Table 23. Verbal axiom items (Q10–Q14): fielded stems, response options, and coding.

Part	Text	Coded as
Q10: Scale invariance		
Question	Suppose we double the 'real income' of each person in a society, when not all the initial incomes are equal.	
Option	After doubling incomes more people have enough money for basic needs, so inequality has fallen.	
Option	Those who had more also get more, so inequality has increased.	
Option	Each person's share remains unchanged, so inequality remains unchanged.	Support
Q11, add variant: Translation invariance		
Question	Suppose we add the same fixed amount to the incomes of each person in a society, when not all the initial income are equal.	
Option	Inequality has fallen because the share of those who had more has fallen.	
Option	Inequality has increased.	
Option	Inequality remains the same.	Support
Q11, deduct variant: Translation invariance		
Question	Suppose instead of adding we deduct a fixed amount from each person's income. Then inequality…	
Option	decreases.	
Option	increases	
Option	is the same	Support
Q12: Population principle		

Part	Text	Coded as
Question	Suppose we replicate a three-person society by merging it with an exact copy of itself (so that we now have a society of six people consisting of three sets of identical twins).	
Option	The income inequality of the six-person community is less than that of the three-person community because in the six-person community there are some people who have the same income.	
Option	The income inequality of the six-person community is greater than that of the three-person community.	
Option	The income inequality of the six-person community is the same as that of the three-person community because the relative income shares remain unchanged.	Support
Q13: Transfer principle		
Question	Suppose we transfer income from a person who has more income to a person who has less, without changing anyone else's income. After the transfer, the person who formerly had more still has more	
Option	Income inequality in this society has fallen.	Support
Option	The relative position of others has also changed as a consequence of this transfer. Therefore we cannot say, a priori, how inequality has changed.	Strongly disagree
Option	Neither of the above.	Disagree
Q14: Decomposability		
Question	Suppose there are two societies, A and B, with the same number of people and with the same total income, but with different distributions of income. Society A is now merged with C, and society B is merged with C' where C and C' are identical.	
Option	The society which had the more unequal income distribution before the merger still has the more unequal distribution after the merger.	Support
Option	We can't say which society has the more unequal income distribution unless we know the exact distributions.	Strongly disagree
Option	Neither of the above.	Disagree

Stems and option wordings as fielded, listed by response code. 'Support' marks the option coded as endorsing the axiom; the 'Strongly disagree' and 'Disagree' labels for Q13 follow Amiel and Cowell's classification. Unlabeled options enter the analysis as non-support.

Appendix G: Reproducibility and session information

This report is computed by the code embedded in its source, making all quantities reproducible.

```
R version 4.6.0 (2026-04-24)
Platform: x86_64-pc-linux-gnu
Running under: Ubuntu 24.04.4 LTS

Matrix products: default
BLAS:   /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3
LAPACK:  /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so;    LAPACK
version 3.12.0

locale:
 [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8        LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8    LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8       LC_NAME=C
 [9] LC_ADDRESS=C               LC_TELEPHONE=C
```

```
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: Etc/UTC
tzcode source: system (glibc)

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] here_1.0.2       lme4_2.0-1       Matrix_1.7-5     tinytable_0.16.0
 [5] scales_1.4.0     ggplot2_4.0.3    stringr_1.6.0    purrr_1.2.2
 [9] tidyr_1.3.2      dplyr_1.2.1

loaded via a namespace (and not attached):
 [1] generics_0.1.4     stringi_1.8.7      lattice_0.22-9     digest_0.6.39
 [5] magrittr_2.0.5     evaluate_1.0.5     grid_4.6.0         RColorBrewer_1.1-3
 [9] fastmap_1.2.0      rprojroot_2.1.1    jsonlite_2.0.0     reformulas_0.4.4
[13] Rdpack_2.6.6       cli_3.6.6          rlang_1.2.0        rbibutils_2.4.1
[17] splines_4.6.0      withr_3.0.3        yaml_2.3.12        otel_0.2.0
[21] tools_4.6.0        nloptr_2.2.1       minqa_1.2.8        boot_1.3-32
[25] vctrs_0.7.3        R6_2.6.1           lifecycle_1.0.5    MASS_7.3-65
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[33] Rcpp_1.1.1-1.1     xfun_0.59          tibble_3.3.1       tidyselect_1.2.1
[37] knitr_1.51         farver_2.1.2       htmltools_0.5.9    nlme_3.1-169
[41] labeling_0.4.3     rmarkdown_2.31     compiler_4.6.0     S7_0.2.2
```

Citing this document

Nataliia Petliak (*2026-10-22*). CODECHECK Certificate 2026-029. CODECHECK Community on Zenodo. https://doi.org/10.5281/zenodo.23666248

About CODECHECK

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```
import session_info2 as si
si.session_info(os=True, cpu=True, gpu=True, dependencies=True)
```

```
pyzmq 27.2.0
debugpy 1.8.22
ipython 9.17.1
platformdirs  4.13.5
jupyter_core  5.9.1
urllib3 2.8.0
tornado 6.6
PyYAML  6.0.3
numpy 2.5.4
ipykernel 7.4.0
pymupdf 2.1
pandas  3.0.6
requests  2.34.2
python-dateutil 3.0.0
traitlets 5.16.1
jupyter_client  8.10.0
comm  0.2.3
typing_extensions 4.16.0
Pygments  2.21.0
certifi 2026.10.19
psutil  7.2.2
session-info2 0.4.2
appnope 1.0.0
packaging 26.3
six 1.17.0
charset-normalizer  3.5.2
idna  3.20
----  ----
Python  3.12.15 (main, Sep 30 2026, 22:04:19) [Clang 21.0.0 (clang-2100.3.34.2)]
OS  macOS-26.6.2-arm64-arm-64bit
CPU 14/14 logical CPU cores, arm
GPU No GPU found
Updated 2026-11-10 12:10
```

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